

Fuse and Sharpen: Cross-Embedding Fusion with Boundary Recovery for Height & Land Cover

Gating four frozen GFMs, then restoring the object edges their embeddings blur away

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Task & Inputs

The task

Per 256×256 tile, predict 4 channels:

[building, vegetation, water, height (m)]

ch 0–2 = binary presence ch 3 = continuous height.

	IoU bld	IoU tree	IoU water	RMSE bld _H	RMSE veg _H
weight	25%	15%	15%	25%	20%

Score = $\sum_i w_i \text{IoU}_i + \sum_j w_j \max(0, 1 - \text{RMSE}_j / X_j)$, $X_{\text{bld}}=3\text{m}$, $X_{\text{veg}}=5\text{m}$.

Everyone gets the *same frozen embeddings*. No raw imagery $\Rightarrow$ a score gap is a **method** gap.

The inputs: two families of embedding

Embedding	Source	Shape	Note
AlphaEarth	multi-sensor	64×256^2	dense, full-res
Tesseract	multi-sensor	128×256^2	dense, full-res
TerraMind S1/S2	SAR / optical	768×16^2	post-encoder tokens
Thor S1/S2	SAR / optical	768×16^2	raw scale $\pm 17k$

Dense, full-res

per-pixel features $\rightarrow$ boundaries & texture (drives IoU).

Coarse tokens (16^2)

$16 \times$ downsampled $\rightarrow$ global semantics, no detail.

This split — dense vs. coarse — dictates the whole design.

The labels (and how the metric reads them)

Target tensor per pixel: $[c_{\text{bld}}, c_{\text{veg}}, c_{\text{water}}, h/30]$

- ch 0–2: **coverage fractions** $c \in [0, 1]$ (a 10 m pixel is rarely pure).
- ch 3: nDSM **height** in metres, normalised by 30.
- `valid_mask`: separate global (nodata) + height (nDSM holes) masks.

How the metric binarises them — and why it matters later:

- **IoU** compares $\text{pred} > 0.5$ vs. official $\text{GT} = \text{coverage} > 0.10$.
- **RMSE** is masked *per class* (only pixels of that class count).

Data observations

Two label defects we found — and masked

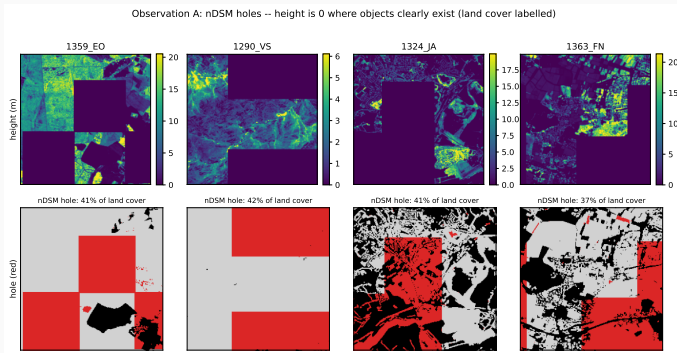
Auditing the GT surfaced two *systematic* defects. Both are handled by the two-channel `valid_mask`, so a defect never becomes a wrong training signal.

Defect	What we see	How we handle it
nDSM holes	height = 0 on labelled objects (often whole sub-tiles)	drop px from <i>height</i> loss; keep presence
Deleted footprints	building label empty where nDSM + embeddings say building	drop px from <i>presence/seg</i> loss; keep height

- `valid_mask` = 2 channels: (0) global/presence, (1) height.
- Each defect zeroes *only* its channel $\Rightarrow$ the other task keeps learning.

Observation A: nDSM holes

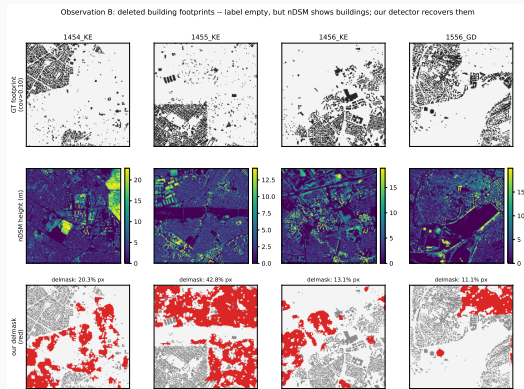
height label



Top: nDSM height as stored (the label writes a literal 0 → dark). Bottom: red = labelled land cover with height = 0 (up to ~40% of px). **Fits:** `ndsm_hole = height==0 & has_landcover` → dropped from height RMSE only; presence kept.

Observation B: deleted building footprints

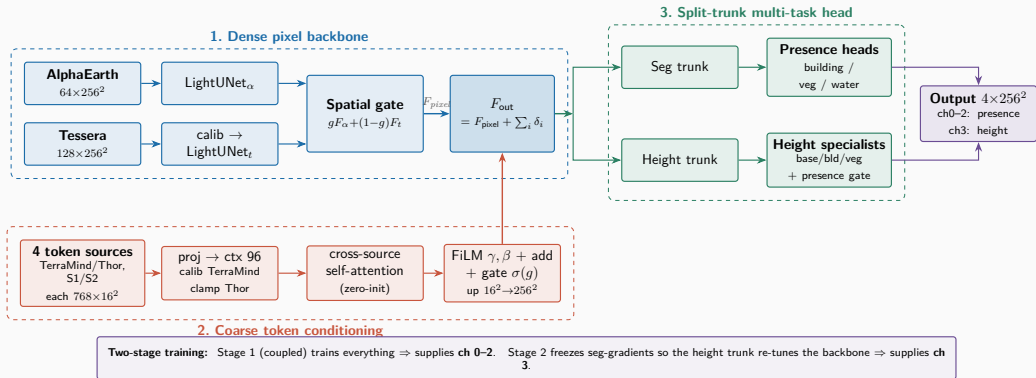
presence label



Footprint empty (top) where the nDSM (mid) shows buildings; a density-gap detector ($\text{blob} \geq 50\text{px}$) recovers the region (red, bottom). **Fits:** dropped from presence/seg loss, height kept — never punish a *correct* building.

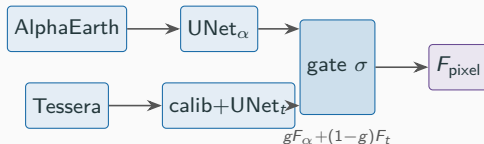
Architecture

Architecture at a glance



■ dense pixel ■ coarse token ■ split head ■ output / two-stage

Design 1 — Dense pixel backbone



- **Fits:** both are dense $256^2 \rightarrow$ a UNet yields sharp, IoU-friendly masks.
- **Fits:** they differ in reliability $\rightarrow$ the gate trusts each per pixel; bias init +4 $\Rightarrow$ starts AlphaEarth-dominant.

ViT / SegFormer / HRNet / UPerNet backbones were all tried and lost.

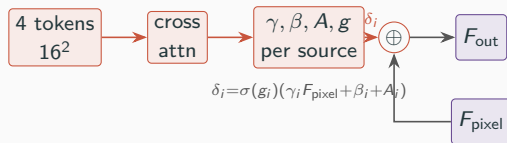
LightUNet (base_ch=48)

- 4 levels: $48 \rightarrow 96 \rightarrow 192 \rightarrow 384$.
- $3\times$ MaxPool down ($256 \rightarrow 32$), bilinear up + skips.
- DoubleConv blocks, BatchNorm.
- `forward_features` $\rightarrow$ 48-ch map at full res.

Tessera entry & gate

- ChannelCalibration: per-channel affine $x \cdot s + b$ (re-normalise the GFM's arbitrary scales).
- Gate = *rich*
conv $\rightarrow$ GN $\rightarrow$ GELU $\rightarrow$ conv, zero-init weights, bias +4 (tied: g and $1-g$).

Design 2 — Coarse token conditioning



- Each source projects to a shared context, cross-attends, then adds a zero-init residual (FiLM + additive + gate).
- Fits:** coarse tokens can't make pixels $\rightarrow$ they *modulate* the dense feature, never replace it.
- Fits:** SAR vs. optical see different physics $\rightarrow$ attention lets them inform each other.

Token fusion internals

Attend (16×16)

- 1×1 proj: $768 \rightarrow$ ctx 96 per source.
- 4-head self-attention over the 4×256 tokens.
- learned modality embedding + 2D positional MLP.
- **zero-init** output proj $\Rightarrow$ identity at start.

Inject (256×256)

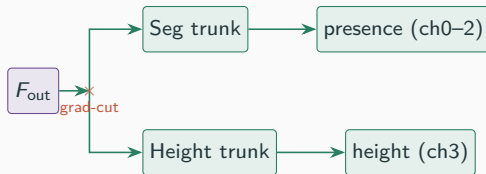
- bilinear up $16^2 \rightarrow 256^2$, then FiLM+add+gate.
- all fusion convs **zero-init**.
- run in fp32 for stability.
- clamps: token/ctx ± 50 , FiLM/add ± 4 .

Respecting each embedding's nature

- **Selective calibration.** Calibrate TerraMind (indices=[0,1]); leave Thor *uncalibrated + clamped*.
Fits: Thor spans $\pm 17k$ — calibrating it collapses water IoU.
- **Zero-init injection.** At init $F_{\text{out}} \equiv F_{\text{pixel}}$.
Fits: boots as a pixel-only net, then *earns* token influence; a weak source degrades gracefully.

Attribution: dropping any of the 4 sources, or any of the 3 pathways, hurt the score.

Design 3 — Split-trunk head



- Two trunks off F_{out} , sharing *no* head weights.
- **Fits:** IoU wants crisp masks, RMSE wants smooth regression — a shared trunk makes their gradients fight; splitting lets each shape its features.
- Presence head *is* the metric (BCE on coverage @ 0.5), not a reparam of fractions.

Head internals & the gradient cut

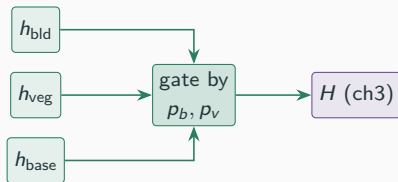
Trunks & heads

- trunk = ConvGNAct + residual + GELU.
- presence `split_all`: independent head per class (`branch_ch` 48, tower depth 2).
- height: 3 independent branches (`hidden` 96, depth 2).

Soft gradient decouple

- forward identity:
 $s t + (1-s) t.\text{detach}()$.
- $s = 1$ coupled, $s = 0$ hard cut.
- this is the knob the two-stage recipe flips.

Presence-gated height specialists

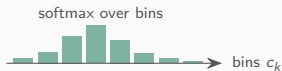


$$H = p_{\text{fg}} (w_b h_{\text{bld}} + w_v h_{\text{veg}}) + (1 - p_{\text{fg}}) h_{\text{base}}, \quad p_{\text{fg}} = 1 - (1 - p_b)(1 - p_v)$$

- **Fits:** the metric masks RMSE *per class* — each specialist is L1-trained, and trusted, only on its own class's pixels.
- Gate uses *alpha* presence logits; each h comes from the soft-bin head (next slide).

Each specialist emits **64 bin logits**, not a scalar; the height is the **soft-argmax** (expectation over bin centers):

$$\hat{h} = \sum_k \text{softmax}(\ell)_k c_k$$

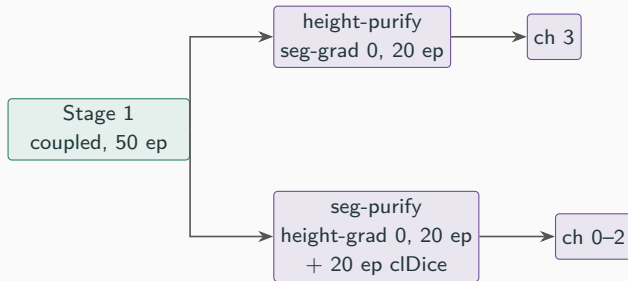


- **One shared grid:** log-spaced $[0, 80]$ m, 64 bins — base/bld/veg share the centers (separate conv projections).
- bin-CE makes the distribution *commit* to a bin rather than collapse to a safe mean (vs. L1-only).
- Non-negative by construction ($c_k \geq 0$).

Refuted: per-class grids (bld $\log[0, 160]$, veg $linear[0, 40]$) — veg soft-argmax blew up ($RMSE_{vH}$ 7.0 vs 2.9), no building gain. Shared grid stays best.

Design 4 — Two-stage training

Two-stage: train, then purify (1 height + 2 seg)



Stage 1 — coupled, 50 ep
AdamW lr 2×10^{-4} ; train everything
jointly $\Rightarrow$ a *shared init* for all purifies.

Stage 2 — purify off stage 1, 20 ep, lr 1.5×10^{-4}

- height-purify (presence-trunk-grad-scale 0) $\rightarrow$ **ch 3**
- seg-purify (height-trunk-grad-scale 0), then +20 ep cIDice on top $\rightarrow$ **ch 0-2**

$\Rightarrow$ the seg-purify *and* cIDice ckpts are both kept (2
decorrelated seg sources $\rightarrow$ 50-member seg).

Loss



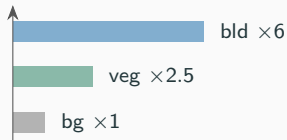
Loss — every term tracks the metric

ImprovedCompositeLoss (presence-centered):

Term	Target	Metric
Presence BCE (ring-weighted)	presence logits	IoU
Presence Tversky (α per class)	presence prob	IoU prec/recall
Height L1 + bin-CE	soft-bin specialists	RMSE
Building-boundary BCE+Dice	aux edge head	IoU edges
Water empty top- k	water logits	water IoU (FP)

Weights: bld height boost $\times 5$, veg $\times 1.5$; bin-CE 0.5; bg weight 0.05; ring $\alpha=2$ (k5); water top- k 512 @ 0.03; boundary 0.25.

Height loss: L1 + bin-CE



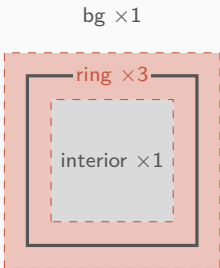
L1 weight $1 + 5m_{\text{bld}} + 1.5m_{\text{veg}}$

Heights normalised /30 m. Two terms:

- **L1** $|\hat{h} - h|$ on the soft-argmax height: blended output (weights, left) + per-class aux on own-class pixels.
- **bin-CE** (0.5): Gaussian soft-target over bins ($\sigma=1.5$) vs. predicted bin distribution → *commit* to the right bin, not collapse to a safe mean. (the key soft-bin term)
- **Base** specialist: bin-CE over *all* valid pixels + blended L1 → bg/water height (≈ 0) is still constrained, just not metric-scored.

Masks use `gt_class>thr` — the same per-class mask the RMSE metric scores on.

Ring-weighted presence BCE



dashed = dilate / erode (k5); solid = mask
edge

- **ring** = $\text{dilate}(m) - \text{erode}(m)$: a $\pm 2\text{px}$ band on the building edge.
- weight $w = 1 + 2 \cdot \text{ring} \Rightarrow$ ring pixels count $3\times$, all else $1\times$ (only the building channel).
- **Fits**: 88% of lost building IoU sits in this band; equal weighting lets the easy interior/bg dominate.
- Reweights the *main* presence logit (what IoU reads), not a separate boundary head.

Inference & Results

- **5-fold, leave-region-out CV** — so local val predicts the public board ($LB \approx$ worst fold).
- **Mean ensemble** across folds; optional 0.3 water-expert blend.
- **Per-class threshold sweep** on OOF, then binarize (bld 0.775 / veg 0.675 / water 0.725).
- **Fits:** IoU is judged at a hard cut, different per class $\rightarrow$ tune each on held-out data.

Ruled out (tried, no gain): TTA, D4 aug, bigger ensembles, region-conditioning.

Final model: xf095_2stage_softbin_covgt10

Two-stage dual-trunk + shared soft-bin. 5-fold (seed 0), purify checkpoints:

fold	Score	iou_bld	iou_wat	RMSE _{bH}	RMSE _{vH}
f0	0.509	0.511	0.455	1.60	2.92
f1	0.541	0.512	0.466	1.40	2.70
f2	0.521	0.497	0.448	1.48	2.78
f3	0.530	0.500	0.465	1.43	2.74
f4	0.525	0.522	0.468	1.48	2.93
mean	0.525				

Worst fold f0 $0.509 \approx$ public LB. Building IoU still boundary-limited (88% of lost IoU in a $\pm 2\text{px}$ band); that is the open lever.

Final Submission

From one model to the final ensemble

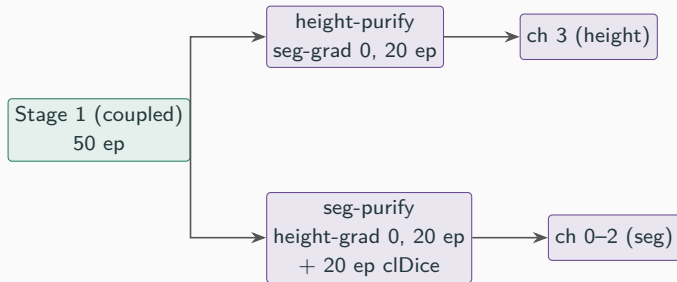
Everything above is *one* model. The submitted system ensembles **5 architectures** $\times$ **5 folds**, each run through **3 training stages**:

Member	pixel_backbone	seeds
U-Net++ (nested decoder)	unetpp	0, 1, 2
UNet 3+ (full-scale skip)	unet3plus	0
TransUNet (attn bottleneck)	unetpp_trans	0

- **Fits:** identical inputs $\Rightarrow$ *architecture* diversity decorrelates errors better than more seeds of one net.
- 5-fold leave-region-out; the ensemble is a plain mean of member-fold probability maps, then thresholded.

U-Net++ was the only backbone that beat plain LightUNet on the official IoU metric; heavier/attention encoders all lost.

Three purify stages off one stage-1



- **height-purify** (presence-trunk-grad-scale 0): freeze the seg side, height re-tunes the backbone $\Rightarrow$ best **ch 3**.
- **seg-purify** (height-trunk-grad-scale 0): the *mirror* — freeze height, seg owns the backbone $\Rightarrow$ sharper **ch 0-2**. **+0.0044 public building IoU**.
- **+cIDice**: a 3rd fine-tune continuing the *seg-purify* ckpt with the cIDice topology loss (next slide) $\Rightarrow$ a 2nd, decorrelated seg source.

clDice — a topology loss for merged buildings

The failure it targets

- Buildings are 1–2 px wide; the model *merges* adjacent ones (predicts 67 components/tile vs. GT's 104).
- **51% of building FP** is inter-building “bridges”; oracle un-merge $\rightarrow$ IoU 0.504 $\rightarrow$ 0.587.

clDice (cl-dice-weight)

- soft-skeleton via iterated min/max-pool; centerline Dice on the building presence map.
- rewards *connectivity*, penalises merges — orthogonal to the area (Tversky) losses.
- added as a 20-ep purify on top of seg-purify.

Ensemble building IoU 0.5039 $\rightarrow$ 0.5072 OOF. **Refuted:** clDice baked into *stage-1* — it pulled the backbone toward topology and *degraded height* ($\text{RMSE}_{bH} + 0.006$). Late-purify clDice keeps height untouched.

Seg from two decorrelated sources (50-member)

Final channels:

- **seg ch 0–2** = mean of **50** member-folds = 25 clDice $\cup$ 25 seg-purify (the two share a stage-1 but diverge in the seg fine-tune $\Rightarrow$ decorrelated).
- **height ch 3** = mean of **25** height-purify member-folds *only* (adding stage-1 height *hurt* — it is strictly worse).

Fits: pure variance reduction — no parameter is tuned on the public board, so it transfers to the private split. On public it lifted all three IoUs (bld +0.0013, veg +0.0011, water +0.0015).

Height calibration: explaining the local→public gap

Local (OOF, tile-RMSE) vs. public, *no shift*:

	OOF tile	OOF pixel	public
RMSE_{bH}	1.48	2.32	1.81
RMSE_{vH}	2.74	2.97	3.08

- **Convention is tile-based:** the tile gap is a *consistent* +0.33 m on *both* heights (region shift); the pixel gap is inconsistent (building would improve — impossible).
- **Mechanism:** soft-argmax *range compression* (tall under-predicted) $\times$ a taller test region $\Rightarrow$ systematic under-prediction.

Height correction: per-class calibration (applied in final)

Building: de-compression (used)

- $h \rightarrow 1.05h + 0.116$ (slope > 1 , un-squeeze the compressed tall tail).
- motivated by the strong compression curve (-4.5 m @ 15 m).
- region-invariant $\Rightarrow$ private-safe. RMSE_{bH} $1.807 \rightarrow 1.782$.

Vegetation: additive shift (used)

- $h \rightarrow h + 0.12$ on predicted-veg pixels (pure mean offset).
- corrects the small public veg bias. RMSE_{vH} $3.079 \rightarrow 3.072$.

Both corrections are region-invariant (a mean offset, plus a mild slope for buildings), not public-fitted; the residual $+0.33$ m is irreducible region-shift scatter.

Final result & trajectory

Submission	Score	bld	veg	wat	RMSE _{bH}	RMSE _{vH}
3-seed U-Net++	0.5018	0.498	0.820	0.521	1.807	3.079
5-member arch ensemble	0.5026	0.498	0.819	0.526	1.804	3.079
+ seg-purify + bld shift	0.5061	0.502	0.821	0.525	1.778	3.079
+ cIDice + veg shift	0.5063	0.502	0.820	0.526	1.778	3.072
+ 50-seg (final)	0.5067	0.503	0.821	0.527	1.782	3.072

- **+0.0049** over the session; the gains are robust (variance reduction + a region-invariant mean shift), not public-fitted.
- Private = best-public auto-selected $\Rightarrow$ near the end only *non-public-tuned* levers were used.

Ablation log — what moved the needle, what didn't

Kept (each +):

- cross-architecture ensemble
- seg-purify (dedicated seg backbone)
- cIDice topology loss
- per-class height calibration (bld de-comp $1.05h+0.116$, veg $+0.12$)
- 50-member dual-source seg
- delmask (drop human-deleted footprints)

Refuted (tried, no gain):

- ViT / SegFormer / HRNet / UPerNet / UNet3+ (solo)
- wavelet down-sampling; deep supervision
- cIDice-from-stage-1
- per-code thresholds; morphology; valley-cut
- distance-watershed (1–2px $\Rightarrow$ degenerate)
- AdaBN; region-conditioning; TTA; D4 aug

Building IoU is at the *embedding-resolution* ceiling (1–2px buildings); the gap to the top teams is a structural model-quality difference (better building-seg / veg-height models on identical embeddings).

Summary

Summary — each design answers “what is the input?”

Design	Driven by...
Dual LightUNet + gate	dense 256^2 , differing reliability
Cross-source token FiLM	coarse 16^2 context (modulate, not pixels)
Selective calib + zero-init	wild scales (Thor); optional correction
Split-trunk head	IoU-crisp vs. RMSE-smooth conflict
Presence-gated height	per-class RMSE mask
Two-stage purify	conflict reaches the backbone
Per-class threshold sweep	IoU judged at a hard per-class cut